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A Bidirectional USB Power Delivery Voltage-Regulating Cable

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***Abstract*— The development of fast charging technology has significantly reduced the charging time for devices. However, with the increase of power level, the thermal and size issues become the limiting factors for fast charging, especially for compact devices. This article presents the concept of the voltage-regulating cable (VRC) which integrates the charging IC and all the associated passive components into the universal serial bus (USB) cable connector. By relocating the power converter and its associated heat from the devices to the cable, this design significantly reduces the device size and provides an alternative solution for the overheating issues. The VRC can regulate a bidirectional step-down voltage for two kinds of typical application scenarios: adapter-to-device (A2D) and device-to-device (D2D) scenarios. In addition, a symmetrical VRC structure is also proposed to facilitate a reversible and user-friendly connection. The designed VRC prototype, fabricated in 180-nm high-voltage BCD process, occupies an area of 4.7 × 2.9 mm^2, including all passive components. It achieves a peak power density of 3.17 W/mm^2 when transferring the maximum power of 42 W in the A2D scenario and a peak power density of 1.36 W/mm^2 with a maximum power of 18 W in the D2D scenario.**

***Index Terms*— Adapter to device (A2D), bidirectional power delivery (BiDi-PD), cable, cable emulator, device to device (D2D), fast charging, hybrid converter, universal serial bus (USB), voltage-regulating cable (VRC).**

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I. Introduction

THE universal serial bus (USB) Type-C power delivery (PD) specification has significantly enhanced USB's capabilities and power handling capacity, leading to its widespread adoption in recent years. Originally designed as a basic data transfer interface, USB has evolved into a universal standard, integrated ubiquitously in various scenarios. With advancements in fast-charging technologies and the USB PD protocol, USB has also become a primary power source for smartphones, laptops, and various other electronic devices [1]. The demand for higher power output capabilities has driven the evolution of USB PD, allowing a single cable to support a range of devices, such as mobile phones, laptops, monitors, and even e-scooters. Higher V_{BUS} voltage settings within USB PD have been particularly beneficial, as they allow for the delivery of higher power levels without proportionally increasing the current in cable, thereby minimizing power losses and alleviating thermal issues. Despite these advances, the increase in power levels still presents new challenges, particularly in managing heat dissipation in compact, portable devices. Therefore, achieving efficient thermal management while maintaining a compact form factor remains a critical concern.

Conventionally, buck–boost switched inductor converters are commonly utilized for fast charging [2], [3]. In these converters, the power inductor is one of the main sources of power loss and also occupies the largest area [3], [4]. To address this problem, an attractive solution is to relocate the power inductor from the high-current output side to the low-current input side [5], [6], [7]. Such inductor-first topologies considerably reduce the average current on the power inductor and thus allow the use of a smaller inductor [8], [9]. Furthermore, moving the inductor to the input side also allows the replacement of the discrete inductor by the parasitic inductor of the USB cable [10], [11], eliminating the need for any on-board magnetics and reducing the on-board power dissipation. For example, the hybrid converters in [11] and [12] reuse the USB cable as the power inductor to generate suitable output voltages for one-cell (1S) and two-cell (2S) series stack batteries. Besides, simultaneous data and power transfer and electromagnetic interference (EMI) tests were conducted to validate the feasibility of reusing the USB cable [12].

However, prior solutions still integrated the power converter chip inside the portable devices. It necessitated that the on-board power converter design has to accommodate various

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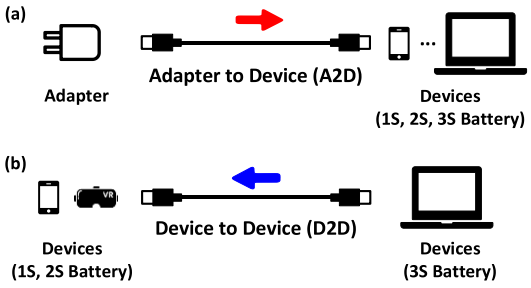


Fig. 1. Two kinds of typical application scenarios (a) A2D and (b) D2D.

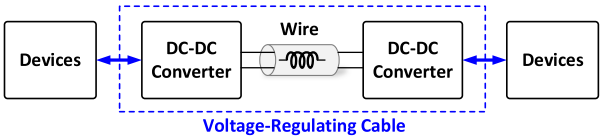


Fig. 2. Structure diagram of the proposed VRC.

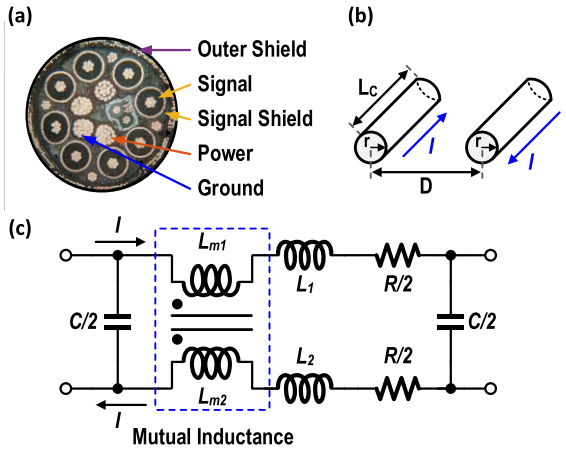


Fig. 3. (a) Cross-sectional view of the USB cable, (b) two-wire single-phase transmission line model, and (c) equivalent circuit model of the USB cable.

cable types and lengths, as cable inductance substantially affects the converter's functionality and performance. Moreover, both the physical space requirements and heat dissipation from the converter remained significant challenges within the device itself.

When integrating the dc–dc converter into the USB cable, multiple application scenarios must be considered, as illustrated in Fig. 1. There are two typical charging scenarios. The adapter-to-device (A2D) forward charging scenario involves different series-stacked batteries (1S, 2S, and 3S) being charged by the wall adapter. Device-to-device (D2D) reverse charging, where a device with a higher battery cell stacked charges another one with a lower stack, is also essential for daily use [13], [14]. For instance, it is desirable to use a notebook (3S) to charge a smartphone or other peripherals (1S or 2S). However, previous inductor-first designs [10], [11], [12], [15] are unable to support such bidirectional step-down PD.

In this article, as illustrated in Fig. 2, we present a reconfigurable bidirectional PD (BiDi-PD) converter to demonstrate the idea of the voltage-regulating cable (VRC) [16]. The VRC not only integrates the charging IC and all the associated passive components into the Type-C connector but also provides bidirectional step-down voltage regulation for both A2D and D2D charging scenarios. By relocating the voltage converter and its associated heat from the devices to the cable, this design significantly reduces the device size and overheating issues. The cable can also serve as a heat sink for the converter chip, significantly alleviating the thermal issue of compact devices. In addition, a symmetrical VRC structure is also proposed to facilitate a reversible and user-friendly connection.

The organization of this article is as follows. Section II presents the impedance analysis and measurement of USB cable and introduces the operation principle of the proposed VRC. Section III gives the system block diagram and the circuit implementation of the cable emulator and reconfigurable bootstrap capacitor circuit. Section IV exhibits the measurement results of the VRC. Finally, Section V draws the conclusions of this work.

II. Proposed VRC

A. Cable Impedance Analysis and Measurement

The cross-sectional view of a widely used USB Type-C cable is presented in Fig. 3(a) [17]. The cable primarily comprises five components: the power wire, ground wire, signal wire, signal shield layer, and outer shield layer. The power and ground wires are responsible for energy transmission, while the signal wires and shielding layers ensure signal transfer without interference from the power wires. Besides, the outer shield layer is also crucial for isolating the USB cable from external environmental factors.

The electrical characteristics of the USB cable can be modeled using a two-wire, single-phase transmission line, as shown in Fig. 3(b). Assuming that the two conductors are perfectly parallel, carry equal and opposite currents, and are separated by D, the total inductance L can be expressed as

$$L = \frac{\mu_0 L_C}{\pi}\left(\ln\frac{D}{r} + \frac{1}{4}\right) \tag{1}$$

where L_C represents the cable length and r denotes the wire radius. The magnetic constant, u_0, which defines the strength of a magnetic field, is given as $4\pi \times 10^{-7}$. In addition, the equivalent lumped circuit model is depicted in Fig. 3(c), where inductors L_1 and L_2 account for the self-inductance effect, while L_{M1} and L_{M2} account for the mutual inductance. Resistor R represents the resistive loss across the cable. Capacitor C models the capacitance between the two wires.

Since the currents in the two conductors are equal in magnitude but flow in opposite directions, the mutual inductance induces an opposing voltage against self-inductance, greatly reducing the overall inductance of the cable. As the distance D between the conductors decreases, the coupling between them becomes stronger, leading to higher mutual inductance and a lower total inductance.

Fig. 4 presents the measured cable impedance across different lengths, brands, shapes, and temperatures. As depicted in Fig. 4(a) and (b), the measurement data show a significant inductance of approximately 150 nH of inductance per meter for USB cables. Compared to discrete inductors,

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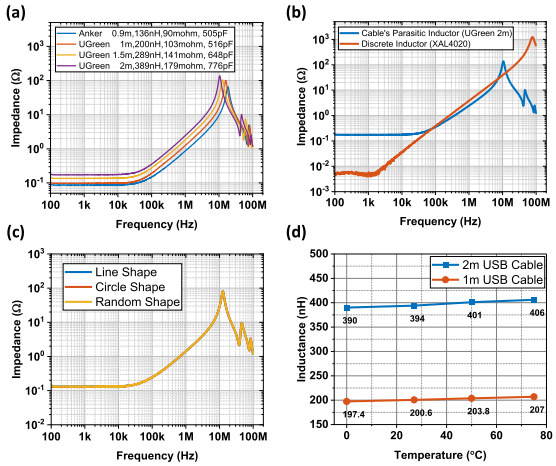


Fig. 4. Measured USB cable impedance across different (a) lengths, brands, (b) types, (c) shapes, and (d) inductance of different temperatures.

cable inductors exhibit higher low-frequency resistance and larger high-frequency capacitance. However, the resistive component does not introduce additional dc energy losses, as it inherently exists in conventional solutions as well. Besides, the capacitive effects only become significant at frequencies above 10 MHz, allowing the cable to be utilized as the power inductor in dc–dc converters operating at switching frequencies below 10 MHz.

The impact of USB cable geometry when bent is another important consideration [10]. Fig. 4(c) compares the cable impedances across different configurations: circular, straight, and irregular (arbitrary) shapes, revealing minimal variation. This geometric insensitivity can be attributed to the outer metallic shielding layer, which effectively contains the magnetic field within the cable [17]. In addition, cable temperature rises during power transfer due to losses. Fig. 4(d) illustrates the measured relationship between inductance and temperature, showing only a 4% increase in inductance as temperature varies from 0 °C to 75 °C. These results demonstrate robust inductance stability across typical operating temperature ranges.

B. Proposed Bidirectional VRC Structure

In this section, we present the working principle of the proposed VRC, as illustrated in Fig. 5. The proposed VRC reuses the USB cable as the power inductor and integrates a BiDi-PD converter into the USB connector to perform the voltage conversion, achieving an extremely compact form factor. The BiDi-PD converter only consists of five integrated power switches S_1–S_5, one flying capacitor C_F, and one filtering capacitor C_L. The implemented VRC exhibits versatility across different application scenarios. Here, we present two typical applications: A2D forward and D2D reverse scenarios [16]. In these configurations, the right cable terminal connects to a battery, while the left terminal interfaces with either a power source or a load.

When the left cable terminal connects to a power source, energy transfers from the source to the battery, as illustrated in Fig. 6 (top left). This configuration corresponds to the scenario where the adapter charges a notebook or other devices. In such

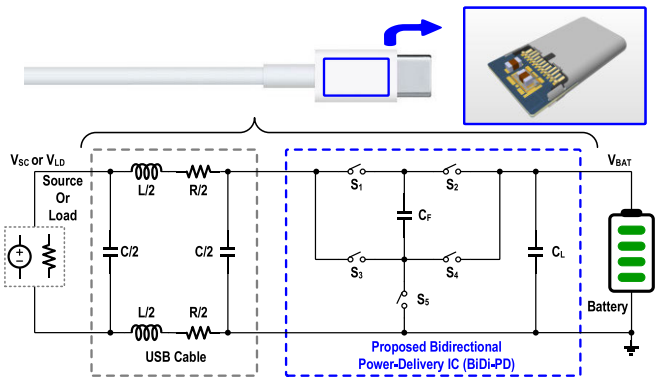


Fig. 5. Proposed VRC structure and bidirectional power-delivery IC (BiDi-PD) circuit diagram.

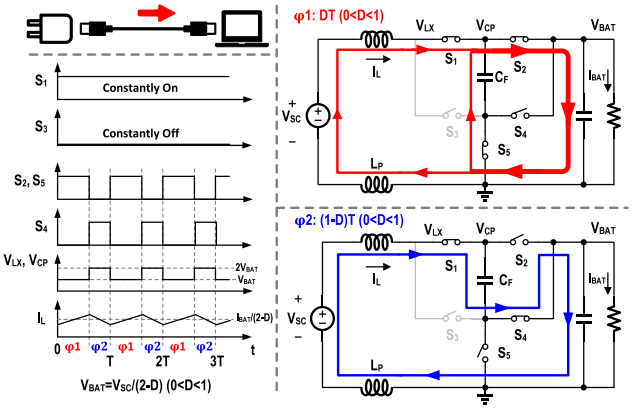


Fig. 6. A2D application scenario (top left), corresponding HPT mode waveforms (bottom left), and operation principle (right).

A2D applications, high-power transfer (HPT) to the battery is usually achieved by fast charging. Accordingly, the BiDi-PD IC operates in the HPT mode, utilizing the cable inductor to carry the input (lower) current of the step-down converter for higher efficiency [18].

The converter operating in HPT mode involves two operating phases with S_1 constant on and S_3 constant off, as illustrated in Fig. 6 (right). During phase φ_1 for a duration of DT, S_2 and S_5 are turned on, with the inductor and flying capacitor supplying power to the battery. In phase φ_2 for a duration of $(1 - D)\ T$, S_4 switches on, allowing the inductor to softly charge C_F and the battery. The attainable voltage conversion ratio (VCR) in this mode can be expressed as

$$\frac{V_{BAT}}{V_{SC}} = \frac{1}{2 - D}(0 < D < 1). \tag{2}$$

On the other hand, when the left cable terminal is connected to the load, as shown in Fig. 7 (top left), energy is transferred from the battery to the load. This configuration corresponds to the scenarios such as a smartphone, or other peripherals being charged by a notebook. In such D2D applications, a step-down converter is required with relatively low power transferred to accessories. However, a wide VCR is necessary due to the varying needs of different applications. Based on these requirements, two feasible solutions can be considered. The first is the buck converter with S_4 constantly off and S_1 and S_3 constantly on. The second topology is a three-level converter

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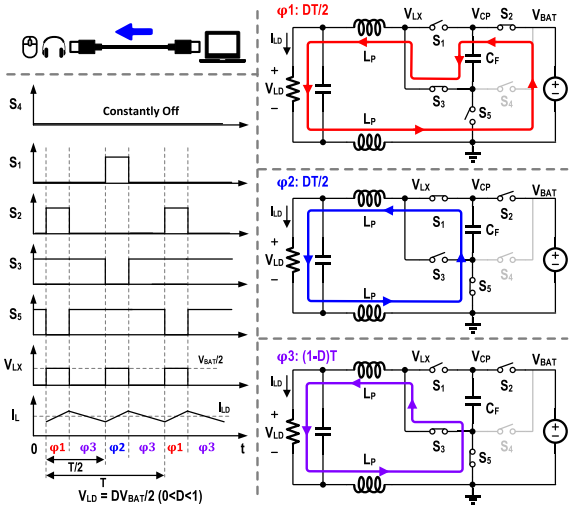


Fig. 7. D2D application scenario (top left), corresponding WCR mode waveforms at 0 < VCR < 0.5 (bottom left), and operation principle at 0 < VCR < 0.5 (right).

with only S_4 constantly off. The three-level converter offers several advantages, including doubled switching frequency, reduced voltage stress, and fully soft-charged flying capacitors. These characteristics help minimize cable inductor current ripple [9] and prevent reverse current flow, ultimately leading to improved conversion efficiency [19]. This mode, also termed wide conversion ratio (WCR), allows for an attainable VCR range from 0 to 1, accommodating various D2D charging scenarios.

The operating principle of WCR mode for VCR ranging from 0 to 0.5 is illustrated in Fig. 7 (right). The converter operates in four phases, with S_4 remaining constantly off. In phase φ_1, which lasts for a time of $DT/2$, the battery energizes the inductor through the flying capacitor. Then, in phase φ_3, the inductor connects to the ground and de-energizes. During phase φ_2, also lasting for $DT/2$, the flying capacitor energizes the inductor, and in the final phase, the inductor connects to the ground and de-energizes once again. The achievable VCR in this operation can be expressed as

$$\frac{V_{\mathrm{LD}}}{V_{\mathrm{BAT}}} = \frac{D}{2} (0 < D < 1). \tag{3}$$

For the conversion ratio ranging from 0.5 to 1, phase φ_3 can be replaced by connecting the inductor to V_{BAT} following a similar approach used in a three-level converter. The attainable VCR in this operation can be expressed as

$$\frac{V_{LD}}{V_{\mathrm{BAT}}} = \frac{1+D}{2} (0 < D < 1). \tag{4}$$

C. Symmetrical Dual-Chip VRC Structure

In this work, we also propose a symmetrical dual-chip VRC structure, as illustrated in Fig. 8, which integrates two BiDi-PD chips into both ends of the cable to enable a reversible and user-friendly connection. This symmetrical VRC design ensures that the cable can be plugged in without worrying about which end to use.

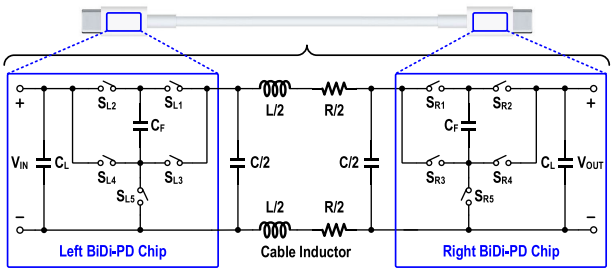


Fig. 8. Proposed symmetrical dual-chip VRC structure and its circuit diagram.

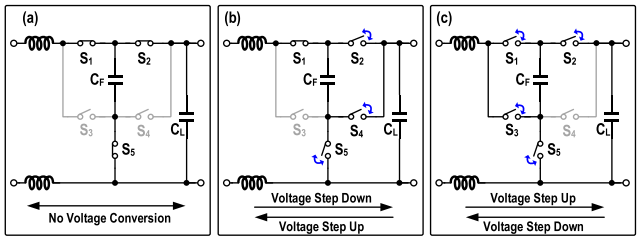


Fig. 9. Three different operation modes of each chip in symmetrical VRC structure. (a) Bypass mode, (b) HPT mode, and (c) WCR mode.

TABLE I
CONFIGURATIONS OF THE SYMMETRICAL VRC

Left Chip Mode	Right Chip Mode	Conversion Type	Conversion Ratio (0<D<1)	Conversion Range	Conversion Efficiency	Application Scenario Example
Bypass	HPT	Buck	1/(2-D)	[0.5, 1]	High	Adapter to Notebook
WCR	Bypass	Buck	D/2 or (1+D)/2	[0, 1]	Medium	Notebook to Mobile Phone
HPT	Bypass	Boost	2-D	[1, 2]	High	Mobile phone to peripheral
Bypass	WCR	Boost	2/D or 2/(1+D)	[1, +∞]	Medium	Mobile phone to Notebook
Bypass	Bypass	N/A	N/A	1	Very High	Compatible with Conventional On-board IC solution

In addition to the HPT and WCR modes mentioned earlier, the chip can also operate in a bypass mode, as shown in Fig. 9(a). In this mode, the chip performs no voltage conversion, with S_1, S_2, and S_5 kept constantly on. The flying capacitor C_{F} and the load capacitor C_{L} are connected together to function as a filtering capacitor.

By utilizing two chips operating in different modes, the symmetrical VRC offers five distinct configurations to accommodate various application scenarios, as detailed in Table I. In these five configurations, one chip is always in bypass mode to simplify control and implementation.

The first two configurations in Table I both perform step-down buck conversions. The first configuration (HPT mode of the right chip) achieves higher conversion efficiency compared to the second configuration (WCR mode of the left chip), due to several key factors: 1) lower inductor dc current; 2) implementation of a dual-path structure; and 3) fewer switched power path switches. However, the first configuration has a limited conversion ratio of 0.5–1. Therefore, due to the differences in VCR and efficiency, these two configurations are suitable for different scenarios. The first configuration is well suited for notebook charging with an adapter, where the adapter voltage can be adjusted by its internal ac–dc converter. Conversely, the second configuration is ideal for scenarios

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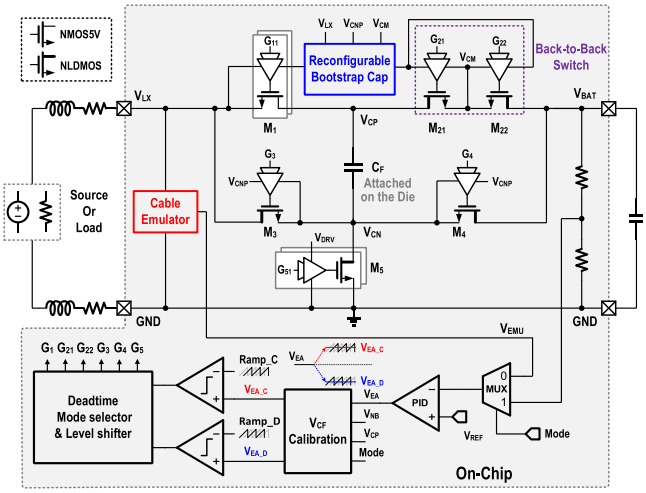


Fig. 10. System block diagram of the proposed VRC IC for bidirectional step-down conversion.

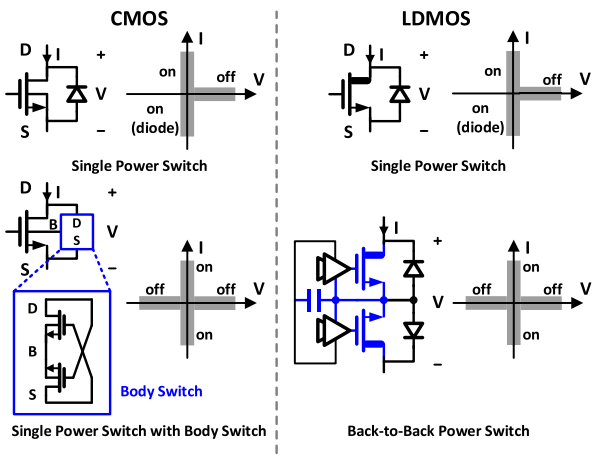


Fig. 11. Four-quadrant power switch implementation for CMOS and LDMOS.

where a notebook serves as a power source to charge other devices.

In addition, the next two configurations perform step-up boost conversions with the exchanged operation modes of the left and right chips of the first two configurations. For example, by swapping the operation modes of the left and right chips in the first configuration, a step-up conversion can be implemented with a reversed conversion ratio

$$\frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}} = 2 - D(0 < D < 1). \tag{5}$$

Similar to step-down buck converters, these two configurations can also be applied in different scenarios. Besides, the last configuration sets two chips both in the bypass mode to ensure compatibility with the conventional on-board charger IC solutions.

III. Circuit Implementations

Fig. 10 shows the system block diagram of the designed BiDi-PD converter. The circuit within the gray dotted box represents the circuits integrated in the chip and it can be divided into two parts: the power stage and the control stage. The power stage comprises six power transistors, the associated bootstrap circuit, and drivers. Among the six power transistors, M_1 is implemented with 5-V NMOS, and the others are implemented with high-voltage lateral diffusion NMOS (NLDMOS).

Due to different transfer power requirements and switched power MOSFETs of the HPT and WCR modes, the optimal sizes of the power switches vary. In the HPT mode, larger sizes of M_1 and M_5 are necessary for HPT, while medium sizes are sufficient in the WCR mode. Therefore, the size of power switch M_1 and M_5 is adjusted based on the operation mode to maintain high conversion efficiency in each mode. Besides, it should be noted the scalable sizing of the switches M_1 and M_5 will also introduce additional capacitive losses due to the increased parasitic C_{DS}. Therefore, the size of these switches must be carefully balanced to tradeoff between conduction losses and switching losses across different modes.

The control stage employs an analog multiplexer (MUX) to select between the cable emulator output V_{EMU} and the battery voltage V_{BAT} for feedback. A shared PID compensation network is utilized to enable stable control loop design [20]. In addition, a calibration circuit is inserted to balance the flying capacitor voltage for the WCR mode operation.

Next, three circuit techniques are implemented in the proposed BiDi-PD converter: 1) a back-to-back switch for bidirectional operation; 2) a cable emulator for obtaining an emulated output voltage for pseudo-feedback loop control; and 3) an area-saving reconfigurable bootstrap circuit.

A. Back-to-Back Power Switch

In this work, the power switch M_2 must handle bidirectional current flow and block the bidirectional voltage, as its OFF-state voltage ranges from $+2V_{\mathrm{BAT}}$ in HPT mode to $-V_{\mathrm{BAT}}/2$ in WCR mode. This kind of switch is also termed a four-quadrant switch when plotting its operation state voltage and current on the coordinate axes, as shown in Fig. 11 [4].

However, a conventional CMOS switch, due to the presence of body diode, cannot block the reverse voltage drop and is thus termed the current-bidirectional two-quadrant switch. Some studies have introduced self-biased body switches to select the body diode connection, thereby achieving the four-quadrant switch [21], [22]. However, this method is not applicable to the high-voltage LDMOS because LDMOS has an inherently asymmetrical structure and fixed body connection [23]. Therefore, in this work, a back-to-back connection structure is employed to implement the switch M_2. In the HPT mode, switch M_{22} is turned on continuously, while M_{21} controls the ON/OFF state of M_2. Similarly, in the WCR mode, M_{21} is turned on continuously, while M_{22} controls the ON/OFF state of M_2. Besides, the sources of M_{21} and M_{22} are connected together to share a single bootstrap capacitor.

B. Cable Emulator Design

In conventional dc–dc converters, the output voltage is fed back to the controller for voltage regulation. However, as depicted in Fig. 12(a), the BiDi-PD converter chip and

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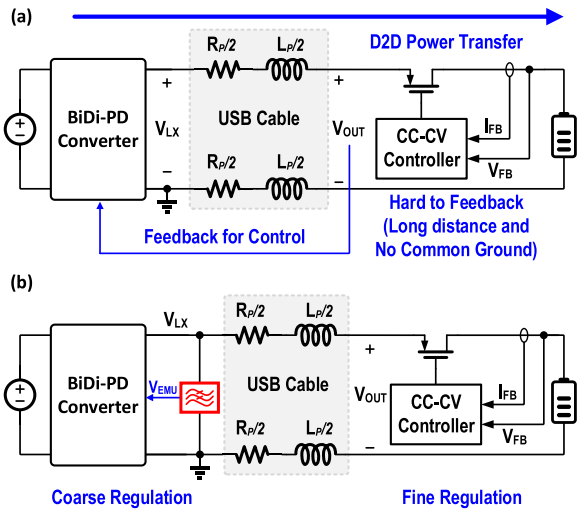


Fig. 12. (a) Separation of the designed chip and load by the USB cable and (b) proposed cable emulator solution.

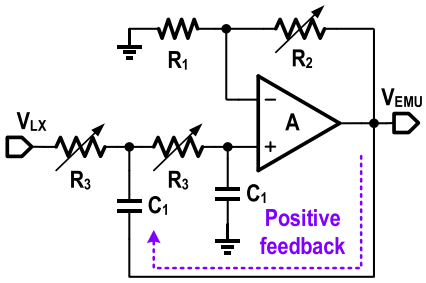


Fig. 13. Circuit diagram of the proposed cable emulator.

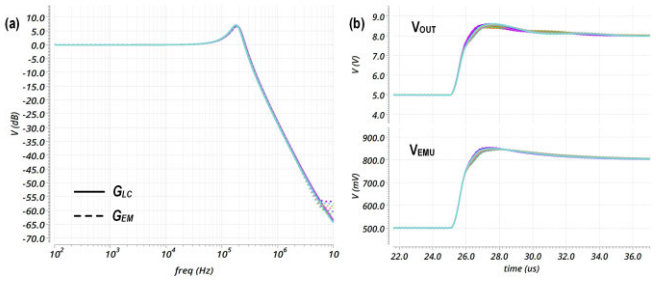


Fig. 14. Simulated results of (a) Bode plot of G_{LC} and G_{EM} and (b) transient waveforms at chip temperature from 0 °C to 80 °C and varied inductance value of 285–315 nH.

load are separated by the USB cable in the D2D application scenario. Due to the long distance and the absence of a common ground between both ends of the cable, directly feeding back the output voltage to the chip side becomes challenging. For instance, utilizing an analog-to-digital converter (ADC) to quantize and feedback the output signal through the USB cable data communication would introduce significant delays in the transmission line as well as substantial costs.

In this work, we introduce an analog circuit-based cable emulator to reproduce the frequency response characteristics of the cable, as depicted in Fig. 12(b). This allows for the coarse regulation of the output voltage by the chip itself without the need to sense the output node. In addition, the existing constant current (CC) and constant voltage (CV) control techniques, along with the associated power switches, can be used to achieve fine regulation of the output voltage and manage the battery charging process [24]. This approach overcomes the challenges of achieving accurate and fast feedback of the output voltage. The system can tolerate certain transient mismatches and dc regulation errors originating from the utilization of the cable emulator. Here, we primarily focus on the design of the cable emulator for a coarse voltage control. The detailed design steps and considerations are given as follows.

First, as depicted in Fig. 12(a), the USB cable transfer function G_{LC} from V_{LX} to V_{OUT} is calculated as

$$G_{LC}(S) = \frac{V_{OUT}}{V_{LX}} \approx \frac{R_O/(R_O + R_P)}{2L_P C_O s^2 + 2\left(\frac{L_P}{R_O} + R_P C_O\right)s + 1} \quad (6)$$

where the parameter R_O and C_O represents the equivalent resistance and capacitance observed at the output side of the circuit. To emulate the frequency response of this transfer function, an operation amplifier-based active second-order filter, also known as tunable Sallen–Key LPF [25] circuit, is employed here, as shown in Fig. 13. The components R_3 and C_1 determine the positions of the two primary poles. Positive feedback is introduced through capacitor C_1 to enhance the voltage gain at the resonant frequency. Furthermore, the components R_1 and R_2 are used to set the dc gain, ensuring that it matches the dc gain of the original transfer function.

The transfer function of the employed cable emulator can be written as

$$G_{EM} = \frac{V_{EMU}}{V_{LX}}(s) = \frac{1 + \frac{R_2}{R_1}}{R_3^2 C_1^2 s^2 + R_3 C_1\left(2 - \frac{R_2}{R_1}\right)s + 1}. \quad (7)$$

Then, associated parameters can be set by comparing the two transfer functions using the equation

$$\begin{cases} R_3 C_1 = \sqrt{2 L_P C_O} \\ \dfrac{R_2}{R_1} = 2 - 2R_P\sqrt{\dfrac{C_O}{2L_P}}. \end{cases} \quad (8)$$

With (8), after specifying the cable inductance L_P and output capacitor C_O, the values of R_1, R_2, R_3, and C_1 can be selected to match the transfer function. To emulate different cables, the resistance values of R_2 and R_3 can be pre-configured in ROM as factory settings. In addition, considering the temperature effect, low-temperature coefficient resistors and capacitors should be utilized here. Fig. 14 shows the simulated Bode plot and transient response with temperature varying from 0 °C to 80 °C and a varying parasitic inductor value (285–315 nH), verifying the effectiveness of the proposed cable emulator across different temperature conditions.

Fig. 15 presents the Bode plot comparison between the tested cable transfer function G_{LC} and the post-simulated emulator transfer function G_{EM}. In the frequency range of interest below 1 MHz, the responses overlap, verifying the above theoretical analysis. Fig. 16 provides the time-domain measurements of the cable emulator. Here, the reference

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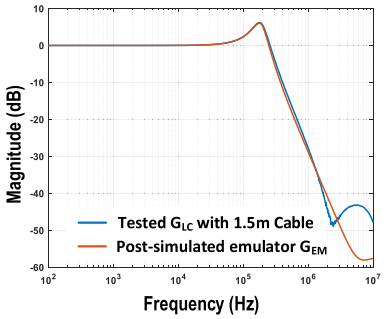


Fig. 15. Bode plot comparison between G_{LC} and G_{EM}.

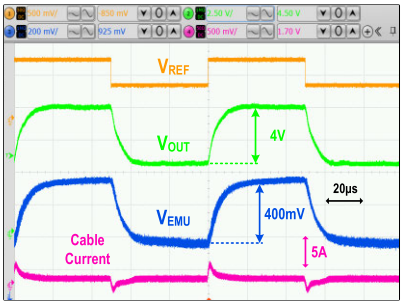


Fig. 16. Time-domain measurements of the cable emulator.

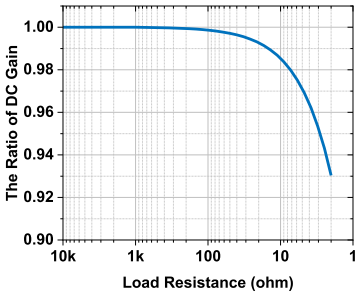


Fig. 17. Ratio of the dc gain of two transfer functions as a function of load resistances.

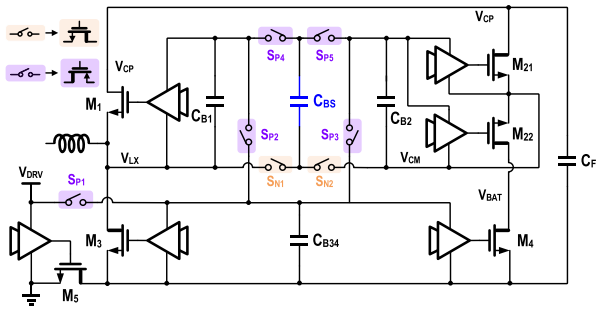


Fig. 18. Circuit implementation of the reconfigurable bootstrap circuit.

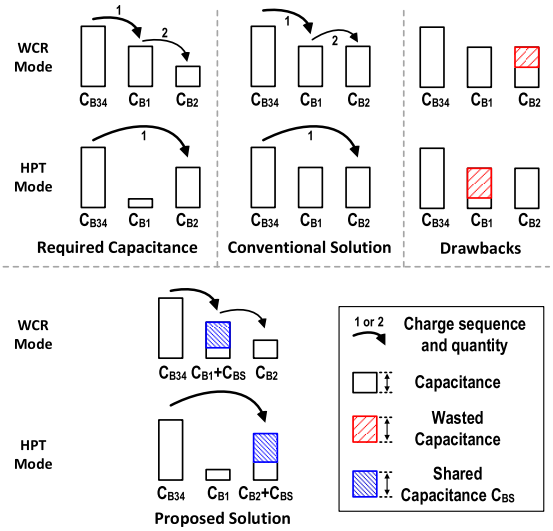


Fig. 19. Comparison of the conventional and proposed bootstrap solutions.

voltage steps from 0.5 to 0.9 V with the emulator output V_{EMU} fed back to the controller. In this closed-loop test, we measured the emulator output V_{EMU} and system output V_{OUT}. The transient waveforms demonstrate that V_{OUT} is well regulated from 5 to 9 V.

Furthermore, due to the existence and large value of R_P, the dc gain of G_{LC} will change with the variation of equivalent output resistance R_O, while the dc gain remains fixed for the cable emulator transfer function G_{EM}. The dc gain ratio of two transfer functions is shown in Fig. 17. When the output resistance is large, this effect can be neglected. As the output resistance decreases, the effect becomes more pronounced and leads to a dc output error (maximum 7% error at the designed lowest resistance). However, this error can be corrected by the finely regulated CC–CV controller and will not affect the battery voltage.

C. Reconfigurable Bootstrap Circuit Implementation

Fig. 18 shows the detailed reconfigurable bootstrap circuit implementation. Each floating power switch requires a bootstrap capacitor connected to the source terminal for proper operation. Since M_3 and M_4 share a common source connection, a single capacitor C_{B34} can be utilized for both switches. The same principle also applies for switches M_{21} and M_{22}. In addition, while conventional cascaded bootstrap circuits can be employed to charge these bootstrap capacitors [19], [26], this traditional approach exhibits limitations regarding the charging sequence and required capacitances in different operating modes: WCR and HPT.

As depicted in Fig. 19, in the WCR mode, C_{B34} needs to provide charge for C_{B1}, and subsequently, except for driving the M_1 switch, C_{B1} also needs to provide charge for C_{B2}, while in the HPT mode, since M_1 does not need to switch, C_{B34} only needs to provide charge for C_{B2}. Thus, only small capacitance is required to maintain the gate voltage of C_{B1}. To be compatible with both modes, the conventional solution makes C_{B1} and C_{B2} both large enough to maintain suitable driver voltages. However, this approach results in certain capacitance waste, as indicated by the red box of Fig. 19.

In this work, a shared bootstrap capacitor C_{BS} is introduced. In the WCR mode, S_{N1} and S_{P4} are constantly on, adding C_{BS} to C_{B1} to provide sufficient capacitance, while in the HPT mode, S_{N2} and S_{P5} are constantly on, incorporating C_{BS} into C_{B2}. By the reconfigurable allocation of C_{BS},

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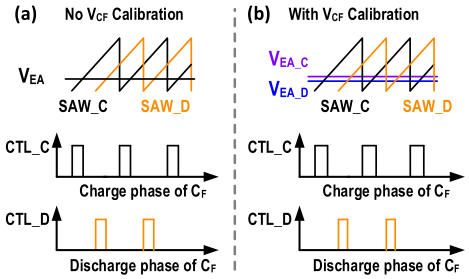


Fig. 20. Generation of control signals of charging phase and discharging phase (a) with calibration circuit and (b) without calibration circuit.

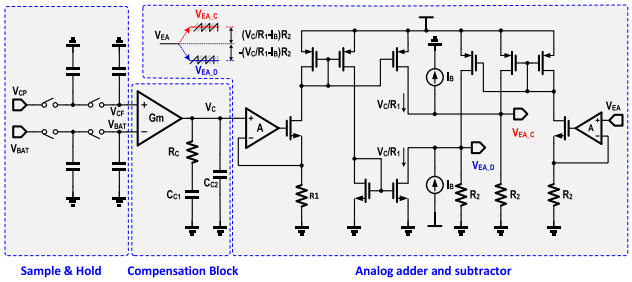


Fig. 21. Circuit implementation of the calibration block of V_{CF}.

this approach achieves the required capacitance values for C_{B1} and C_{B2} with much smaller silicon area.

The quantitative analysis of required bootstrap capacitance is presented in the following. Assume that the driving voltage should exceed $0.9V_{DRV}$ to maintain low switch conduction resistance and each capacitor experiences the same voltage drops during charging and discharging cycles. Based on these assumptions, we can estimate the required bootstrap capacitance in the WCR mode as follows:

$$\begin{cases} C_{B2_WCR} = 30C_{GG_M22} \\ C_{B1_WCR} = 30(C_{GG_M22} + C_{GG_M1}) \\ C_{B34_WCR} = 30(C_{GG_M3} + C_{GG_M22} + C_{GG_M1}) \end{cases} \tag{9}$$

where C_{GG} represents the equivalent gate capacitance. Similarly, in the HPT mode, the required bootstrap capacitance can be estimated as

$$\begin{cases} C_{B1_HPT} = 5C_{GG_M1} \\ C_{B2_HPT} = 20C_{GG_M21} \\ C_{B34_HPT} = 20(C_{GG_M4} + C_{GG_M21}) \end{cases} \tag{10}$$

where C_{B1_HPT} is selected to be only five times the gate capacitance, which is sufficient to maintain M_1 in its ON state. The reduction in bootstrap capacitance can be quantified as the difference between capacitance requirements of two modes.

D. Flying Capacitor Voltage Calibration

In the WCR mode, the power stage operates as a three-level converter, enabling reduced cable current ripple for improved conversion efficiency. However, these advantages are only realized when the flying capacitor voltage remains at $V_{BAT}/2$, which is influenced by practical factors including parasitic capacitance and timing mismatches [27], [28]. Therefore, active balancing techniques are essential.

Conventional three-level converter control involves generating two control signals by comparing the V_{EA} signal with two 180° phase-shifted sawtooth signals for both charging and discharging phases, as shown in Fig. 20(a). However, time mismatches are inevitable in real-circuitry implementations. If the charging phase is longer than the discharging one, V_{CF} increases. In addition, the bottom plate parasitic capacitance also influences the balanced voltage of C_F.

In this work, we split the V_{EA} signal into two signals to generate different duty cycles for controlling the charging and discharging phases of the converter, as illustrated in Fig. 20(b). For example, if V_{CF} is higher than $V_{BAT}/2$, the charging phase will be shortened and the discharging phase will be lengthened to compensate for the parasitic effect.

Fig. 21 presents the implementation of the calibration circuits. First, the flying capacitor voltage (V_{CF}) and battery voltage (V_{BAT}) are acquired through a sample hold circuit. Then, the error signal between two voltages is amplified and compensated to ensure loop stability [29]. Next, the generated signal V_C is superimposed on the V_{EA} signal to generate V_{EA_C} and V_{EA_D}, splitting the V_{EA} signal into two control signals. The relationships can be written as

$$\begin{cases} V_{EA_C} = V_{EA} + (V_C/R_1 - I_B)R_2 \\ V_{EA_D} = V_{EA} - (V_C/R_1 - I_B)R_2 \end{cases} \tag{11}$$

where the bias current I_B is introduced to determine the dc value of the compensation block output V_C. If no calibration is needed to split V_{EA}, V_C can be expressed as

$$V_C = I_B R_1 \tag{12}$$

with the introduction of I_B, the value of V_{EA_C} can be either larger or smaller than V_{EA}, addressing various cases of unbalanced voltage.

Fig. 22 presents the circuit implementations of the out-of-phase sawtooth signals and their associated waveforms. A 50% duty cycle clock signal (CLK) is utilized to generate two reset signals, CLK1 and CLK2, based on the rising edge and falling edges, respectively. These two reset signals are then used to reset the sawtooth signals, SAW_C and SAW_D, with exactly a 180° phase difference. Fig. 23 shows the measurement results illustrating the calibration effect of the designed calibration block. We compared the steady-state waveforms of the WCR mode with the calibration block disabled and enabled. With calibration enabled, the V_{CF} voltage is balanced to $V_{BAT}/2$, and the cable current ripple is reduced.

IV. Measurement Results

The proposed BiDi-PD converter, fabricated in a 0.18-μm BCD process, occupies a chip area of 2.8 × 2 mm, as shown in Fig. 24. The V_{CN} and V_{CP} pad arrays are positioned at the center of the chip for soldering the flying capacitor C_F. The bootstrap capacitors are placed beneath these pads to optimize space utilization. The converter switches at 5 MHz with only two 0603 10-μF capacitors (the nominal value before dc derating). The assembled VRC type-C connector only occupies an area of 16.1 × 8.3 mm, as shown in Fig. 25.

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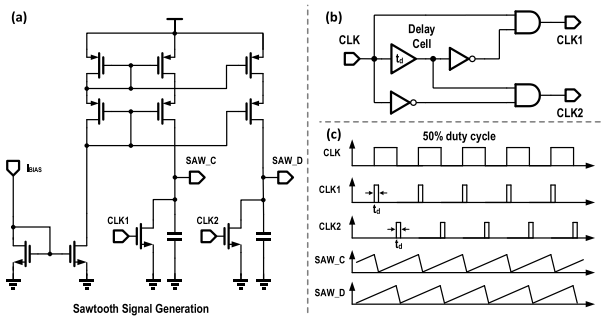


Fig. 22. Circuit implementation of (a) out-of-phase sawtooth signals generation, (b) reset clock signal generation, and (c) associated current waveforms.

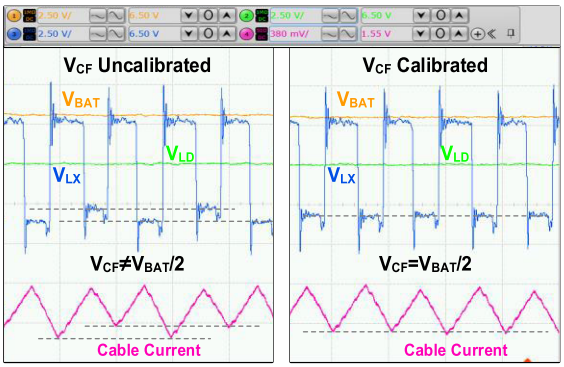


Fig. 23. Measured steady-state waveforms with calibration block disabled and enabled.

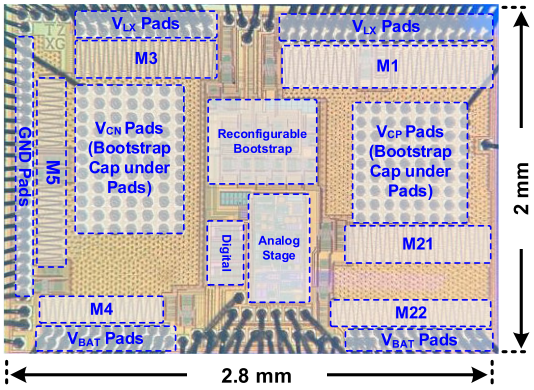


Fig. 24. Chip micrograph of the proposed BiDi-PD converter.

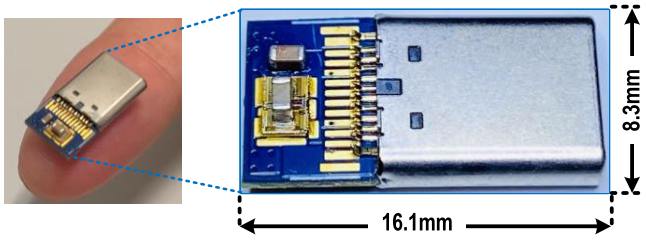


Fig. 25. Assembled VRC connector.

Due to the minimal off-chip passive components used, the designed converter is compact enough to be integrated into the standard USB cable.

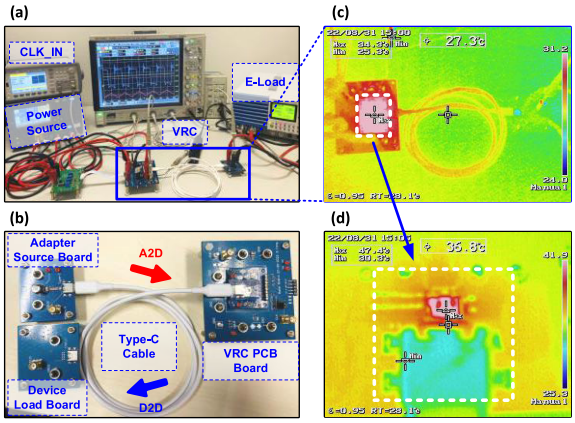


Fig. 26. (a) Measurement environment, (b) detailed test setup in two application scenarios, (c) thermal image of the VRC, and (d) zoomed-in thermal image of the BiDi-PD chip.

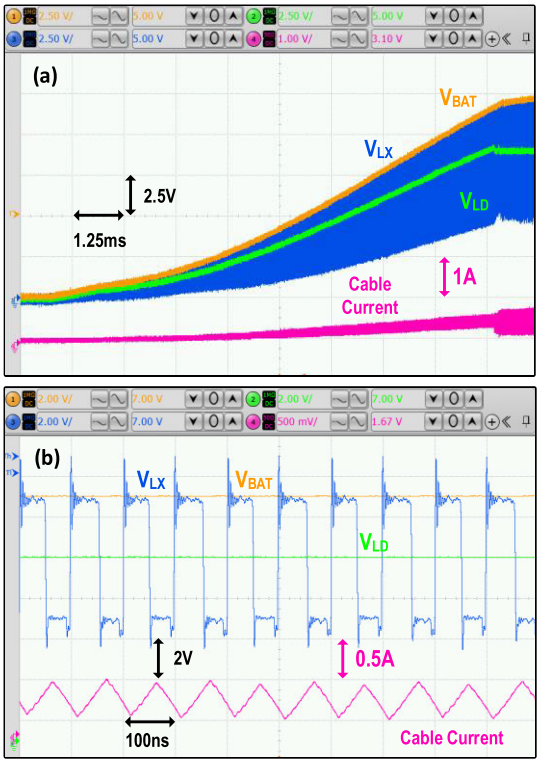


Fig. 27. Measured (a) start-up waveform for HPT mode and (b) steady-state waveform for HPT mode @16–12-V and 1-A output current.

Fig. 26(a) shows the practical measurement environment, with the power supplied by a voltage source. The waveform generator provides the necessary clock signal to the chip. An electronic load is utilized to adjust the load resistance. The core region is denoted by the blue box in Fig. 26(a). The cable transfers power and converts the voltage from one end to the other, with the cable current measured via a current probe on a small cut section. Fig. 26(b) presents the detailed test setup in two application scenarios. Fig. 26(c) and (d) shows the thermal images of the overall system and BiDi-PD chip, with the converter IC being the hottest part, further motivating integration of converter chip outside of devices. In addition, it is evident that the heat loss associated with the inductor

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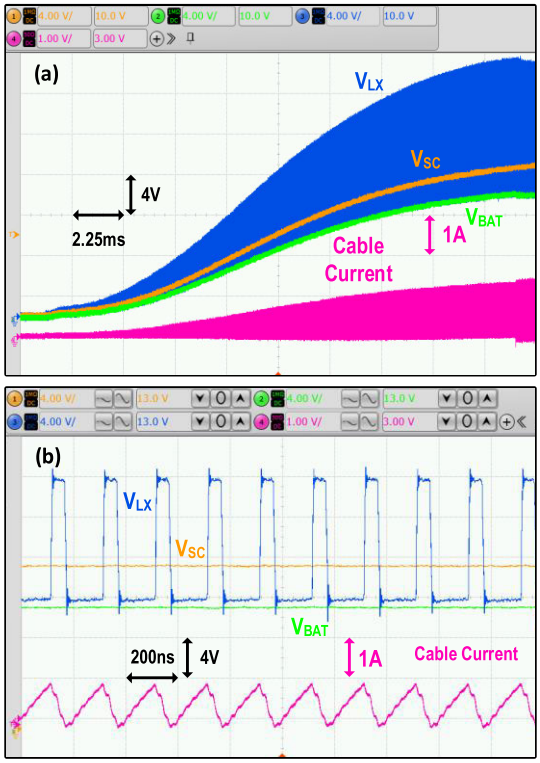


Fig. 28. Measured (a) start-up waveform for WCR mode and (b) steady-state waveform for WCR mode @12–9-V and 1-A output current.

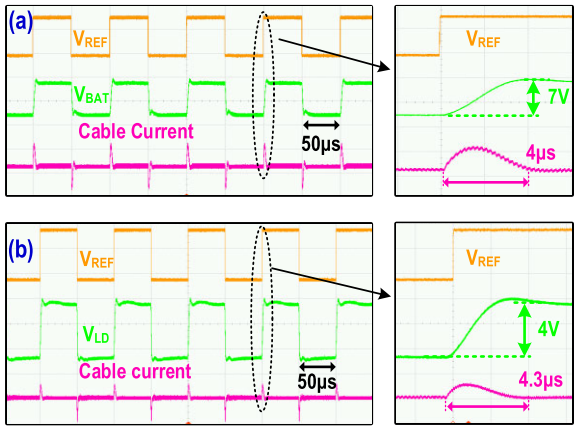


Fig. 29. Measured dynamic voltage scaling waveforms for (a) HPT mode and (b) WCR mode.

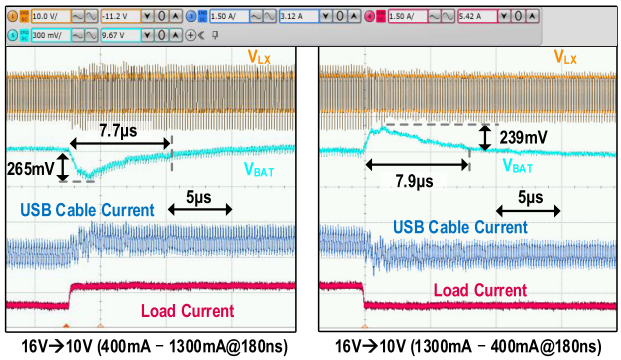


Fig. 30. Measured load transient response in the HPT mode.

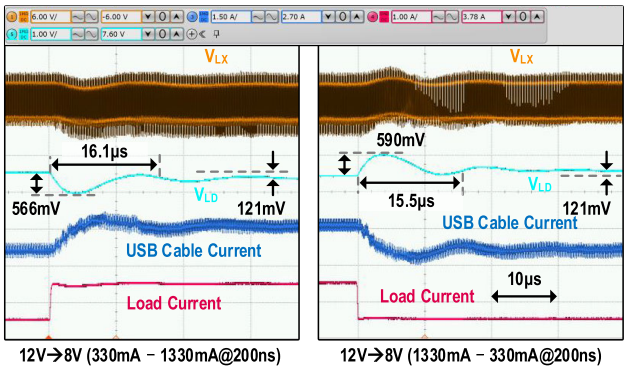


Fig. 31. Measured load transient response in the WCR mode.

(DCR loss and ac loss) is dissipated through the cable, making the cable to become hotter than the surrounding environment. The cable's large surface area in contact with air allows it to dissipate heat from the chip more effectively, particularly for the VRC, which integrates the converter chip into the USB cable. Besides, if heat-dissipating materials are incorporated into the cable, this effect can be further enhanced.

Fig. 27 presents the measured start-up and steady-state waveforms of HPT mode (2-m Type-C cable). In this mode, when the source voltage V_{SC} steps up to 16 V within 22.5 ms, the battery voltage V_{BAT} gradually rises to 12 V accordingly. The switching node V_{LX} switches between V_{BAT} and $2V_{BAT}$ within 200 ns, as expected by its operation principle. Also, the measured cable current waveform, depicted in the pink line, exhibits high-frequency ringing caused by its parasitic capacitances.

Similarly, Fig. 28 presents the measured start-up and steady-state waveforms of WCR (1.5-m Type-C cable) mode. In the WCR mode, as the notebook voltage V_{BAT} steps up to 12 V, switching node V_{LX} switches between V_{BAT} and $V_{BAT}/2$ within 100 ns. The tested waveforms demonstrate the smooth start-up process and normal steady-state operation of the proposed VRC.

Fig. 29 presents the dynamic voltage scaling waveforms of the two operation modes. In the HPT mode, it takes 4 μs for the battery voltage V_{BAT} to step up by 7 V. Also, in the WCR mode, it takes 4.3 μs for the load voltage V_{LD} to step up by 4 V. These two waveforms demonstrate the dynamic regulation capability of the designed prototype.

Fig. 30 shows the load transient response in HPT mode with output current stepping up and down between 400 and 1300 mA within 180 ns, measured with a 2-m Type-C cable. The converter takes 7.7 and 7.9 μs to recover.

Fig. 31 shows the load transient response in the WCR mode with output current stepping up and down between 330 and 1330 mA within 200 ns, measured with a 2-m Type-C cable. The converter takes 16.1 and 15.5 μs to recover. In addition, comparing the light load and heavy load output voltages, there is a 121-mV dc output error as described above.

Fig. 32 gives the conversion efficiency in the HPT mode with voltage stepped down from 16 to 12 V and from 20 to 12 V, tested with a 2-m Type-C cable and an output current (I_{OUT}) ranging from 0.2 to 3.5 A. The system efficiency, including the cable loss, achieves a peak efficiency of 92.8%. By deducting the cable resistance loss using the

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TABLE II
PERFORMANCE SUMMARY AND COMPARISON WITH PRIOR ARTS

	ISSCC'19	TIE'19	ISSCC'21	ISSCC'24	TI BQ24295		This work	
Converter In Cable	NO	NO	NO	No	NO		**YES**	
Conversion Direction	Unidirectional	Unidirectional	Unidirectional	Unidirectional	Bidirectional		**Bidirectional**	
Conversion Type	Step-Down	Step-Down	Step-Down	BuckBoost	Forward mode	Reverse Mode	**HPT Mode**	**WCR Mode**
					Step-Down	Step-Up	**Step-Down**	**Step-Down**
Input Voltage (V)	9	5	1.8	2.7 - 4.2	3.9-6.2	2.7-4.2	**5-20**	**4-12.6**
Output Voltage (V)	6-8	3.3-3.9	0.3-1.5	3.4	2.7-4.2	4.55-5.5	**3.7-13.5**	**3.7-9**
Inductor Location	Input	Input	Input	Middle	Output	Input	**Input**	**Output**
Peak Output Power	31.9W	15.2W	4.13W	4.2W	12.6W	8.25W	**42W**	**18W**
Swithing Frequency	1.8-2.3MHz	2MHz	3MHz	1MHz	1.5MHz	1.5MHz	**5MHz**	**5MHz**
Peak Efficiency (PE)	93.5%	N.A.	N.A.	N.A.	N.A.	N.A.	**92.8%**	**92.4%**
Peak Chip Efficiency **	97.60%	89.70%	98.20%	96.90%	93.20%	93%	**95.1%**	**93.7%**
Power Density* @PE [W/mm2]**	0.32	0.15*	0.11	0.43	0.06*	0.04*	**1.90**	**0.41**
Peak Power Density* [W/mm^2]**	0.69	0.28*	0.72	1.3	0.35*	0.23*	**3.17**	**1.36**

* estimated value according to paper or datasheet, ** Calculated value without cable loss
*** including chip and passive components area

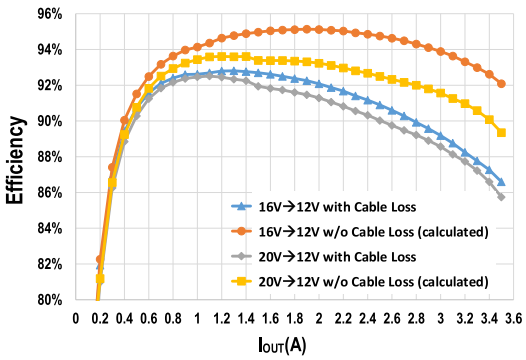


Fig. 32. Measured conversion efficiency in the HPT mode.

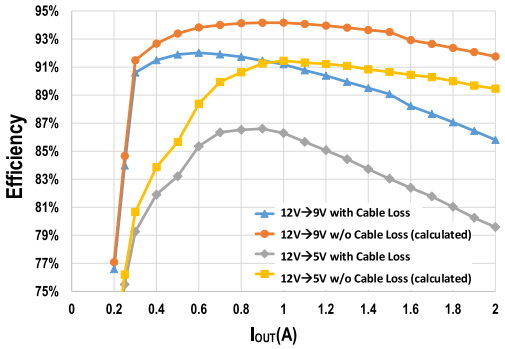


Fig. 33. Measured conversion efficiency in the WCR mode.

equation $I_{\text{rms}}^2 R_{\text{P}}$, the peak chip efficiency, excluding the cable losses, is estimated to be 95.1% at 22.8 W. In fact, the cable loss would be larger than this estimation due to the missing high-frequency loss. However, it is hard to accurately test the cable losses. The peak power density of the HPT mode is as high as 3.17 W/mm^2 due to fewer off-chip components with a maximum power of 42 W.

Fig. 33 presents the conversion efficiency of the WCR mode with voltage stepped down from 12 to 9 V and from 12 to 5 V, which is tested with a 1.5-m Type-C cable with I_{OUT} ranging from 0.2 to 2 A. The achieved peak chip efficiency without the cable loss is 93.7% at 9 W. The peak power density of the reverse mode is 1.36 W/mm^2 with a maximum power of 18 W.

Table II summarizes and compares the performance of the proposed BiDi-PD converter with state-of-the-art designs [6], [10], [15], [30], [31]. This is the first work that integrates the charging converter into the cable connector. Also, it features a bidirectional step-down voltage conversion for different applications. The BiDi-PD converter not only obtains the highest power density but also can transfer a maximum of 42 W to notebooks with a peak chip efficiency of 95.1%. Besides, a peak chip efficiency of 93.7% and a peak power density of 1.36 W/mm^2 is achieved in the WCR mode.

V. CONCLUSION AND DISCUSSION

Ever since the first adoption of USB cable for wired connections, its passive nature persisted over time. However, as fast charging protocols emerge, relying solely on portable devices for voltage conversion has proven to be increasingly challenging and inefficient. In this work, we propose and demonstrate the idea of the VRC with integrated power conversion functionality within the cable connector. The proposed

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VRC eliminates the need for bulky dc–dc converters and the associated passive components in portable devices while also mitigating thermal management challenges. The VRC concept can also be considered as a prototype for the third-generation USB cable, building upon the first generation, which featured a straightforward direct connection between devices, and the second generation, which incorporated electronic marker (E-marker) chips in the USB connector to support diverse information transfer requirements [32].

Furthermore, the application of the third-generation USB cable can be extended to other power transmission areas such as the charging of electrical vehicles (EVs) [33], long-distance high-voltage power transmission [34], and the home power network. By reutilizing the power line as the inductor and integrating the converters into the line, a smarter, more efficient and smaller form factor power transmission network can be expected [35].

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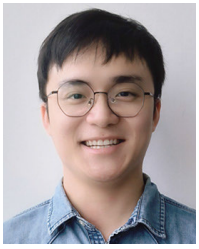
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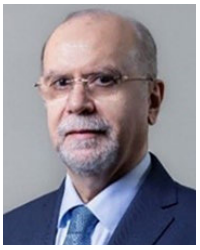
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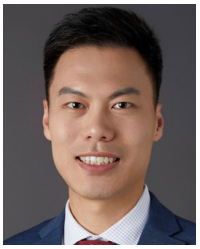
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